

Decoupling the Prediction Objective for Inductive Temporal Link Prediction

Duong Viet Hoang*
Department of Computer Science
Da-Yeh University, Taiwan
Hoangduong4316@icloud.com
*Corresponding author

Duong Viet Huy
Department of Computer Science
Kent State University, United States
huydv6210@gmail.com

Lun-Min Shih
Department of Computer Science
Da-Yeh University, Taiwan
Lmshih@mail.dyu.edu.tw

Abstract—This paper examines a narrow design choice in temporal link prediction: whether the prediction loss should update the temporal backbone. We compare a coupled arm with a gradient-decoupled arm while holding the registered data, split policy, negative sampling, training schedule, and evaluation endpoint fixed. Across CoEdit, MOOC, and Wikipedia, the decoupled arm has the higher mean inductive average precision in the selected runs. The result is a within-protocol observation. It neither establishes causality nor supports a claim against external implementations.

Index Terms—temporal graphs, link prediction, inductive evaluation, gradient decoupling, research reproducibility

I. INTRODUCTION

Temporal link predictors learn from ordered interactions and score candidate edges at a later time. In an inductive split, at least one endpoint of a test edge was absent from training. The setting therefore rewards representations that transfer beyond the identities observed during optimization.

Most implementations let the link-prediction objective update every trainable component. That choice is convenient, but it is not neutral: the prediction head can shape the representation around shortcuts that are useful only for the training population. We study a direct alternative. The decoupled arm stops the score gradient at the backbone boundary, whereas the coupled arm allows it to pass. The comparison changes that route and leaves the registered evaluation protocol unchanged.

The contribution is deliberately bounded. We report the observed ordering of the two registered arms on the current corpora and preserve the complete chain from each table entry to a frozen aggregate, execution job, and checksum. We do not use local proxy baselines to make external performance claims.

II. RELATED WORK

Memory-based temporal models such as JODIE, DyRep, and TGN maintain evolving node state, while attention and mixing architectures such as TGAT and GraphMixer construct representations from temporal neighborhoods [1], [2], [3], [4], [5]. These families motivate careful evaluation under inductive splits because node identity and interaction history are not equally available at test time.

Stop-gradient operators and frozen representations have been studied as optimization and transfer devices in other

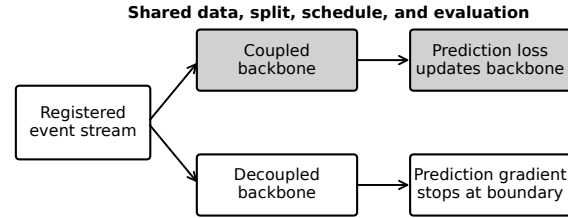


Fig. 1. The registered comparison changes the prediction-gradient route. Data handling and evaluation remain shared.

learning settings [6], [7]. Here they serve a more limited purpose: they define a controlled arm in a registered temporal-link experiment. The present evidence does not establish that the result transfers to other architectures or protocols.

III. REGISTERED COMPARISON

Both arms use the same temporal backbone, corpus identities, chronological split, negative-sampling policy, optimizer configuration, and metric implementation. In the coupled arm, the link loss updates the backbone. In the decoupled arm, the score path receives a detached backbone representation. Figure 1 shows this single experimental contrast.

The primary endpoint is inductive average precision. Each aggregate is formed from the selected seed set, and dispersion is the sample standard deviation. Runs outside the reconciled task matrix, including failed or superseded attempts, cannot enter the aggregate silently.

IV. RESULTS

The decoupled mean is higher for every corpus in Table I. The separation is largest on CoEdit and smaller on the two standard interaction graphs. Figure 2 presents the same ordering without introducing additional reported values.

These observations do not identify a universal mechanism. In particular, the current release does not support claims of irreversibility, superiority to external baselines, hard-negative robustness, or state-of-the-art performance. Those questions require their own registered implementations and evidence.

TABLE I
INDUCTIVE AVERAGE PRECISION IN THE REGISTERED COMPARISON.
VALUES ARE MEAN $\pm$ SAMPLE STANDARD DEVIATION OVER THE
SELECTED SEEDS.

Corpus	Training arm	Mean	SD
CoEdit	Coupled	0.9787	0.0021
CoEdit	Decoupled	0.9891	0.0016
MOOC	Coupled	0.9910	0.0047
MOOC	Decoupled	0.9947	0.0007
Wikipedia	Coupled	0.9967	0.0003
Wikipedia	Decoupled	0.9983	0.0003

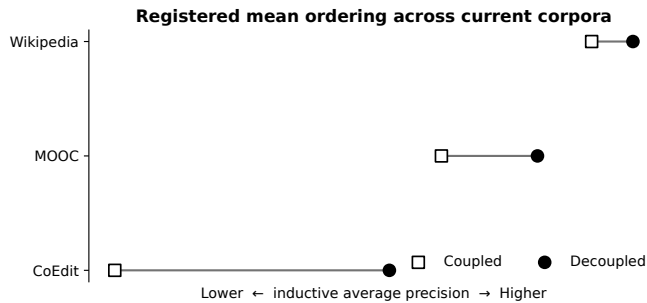


Fig. 2. Ordering of the registered inductive-AP means. Axis ticks and value labels are omitted because the audited values are reported in Table I.

V. EVIDENCE BOUNDARY

The paper is built from an immutable evidence release. Each numeric occurrence in Table I resolves through an explicit selector to the frozen aggregate produced by the registered scheduler reconciliation job. The release also retains the protocol, resolved configuration, dependency lock, dataset identities, environment record, attempt ledger, and final scheduler accounting. The snapshot builder rejects a missing selector, a stale source line, a checksum mismatch, or a value that does not reproduce the printed rounding.

Dataset bytes are not distributed with the source repository. Acquisition is fetch-only and requires the registered upstream digest. This boundary keeps software licensing separate from dataset rights.

VI. LIMITATIONS

The comparison covers one implementation, one protocol, and the current registered corpora. It does not isolate which property of a corpus determines the observed gap. It also does not compare against verified upstream baseline implementations. The result should therefore be read as a controlled within-system measurement rather than a general ranking of temporal graph models.

VII. CONCLUSION

Under the registered inductive protocol, the gradient-decoupled arm has a higher mean average precision than the coupled arm on each current corpus. The evidence supports that descriptive statement and no broader one. The release and conference snapshot preserve the source, execution identity,

and numeric provenance needed to inspect the comparison independently.

REFERENCES

- [1] S. Kumar, X. Zhang, and J. Leskovec, “Predicting dynamic embedding trajectory in temporal interaction networks,” in *Proc. KDD*, 2019.
- [2] R. Trivedi, M. Farajtabar, P. Biswal, and H. Zha, “DyRep: Learning representations over dynamic graphs,” in *Proc. ICLR*, 2019.
- [3] E. Rossi, B. Chamberlain, F. Frasca, D. Eynard, F. Monti, and M. Bronstein, “Temporal graph networks for deep learning,” 2020.
- [4] D. Xu, C. Ruan, E. K rpeo lu, S. Kumar, and K. Achan, “Inductive representation learning on temporal graphs,” in *Proc. ICLR*, 2020.
- [5] W. Cong, S. Zhang, J. Kang, B. Yuan, H. Wu, X. Zhou, H. Tong, and M. Mahdavi, “Do we really need complicated model architectures for temporal networks?” in *Proc. ICLR*, 2023.
- [6] X. Chen and K. He, “Exploring simple Siamese representation learning,” in *Proc. CVPR*, 2021.
- [7] G. Alain and Y. Bengio, “Understanding intermediate layers using linear classifier probes,” 2017.